

Ananta Narayan Shrestha

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Summary

Software engineer with 6+ years of experience, including 4 years shipping FDA-regulated software at Philips Ultrasound R&D — where I was the sole developer of a real-time audio module for the Lumify iOS app (Swift, C++, AVFoundation/AVAudioEngine), a clinical ultrasound product on the App Store — owning it from requirements through delivery. Deep systems background in C++/Rust (real-time, concurrency, performance), suited to iOS roles in media, audio/video, or performance-critical native code. Open to relocation.

Experience

Research Software Engineer Aug 2023 – Sep 2025
National Renewable Energy Laboratory Colorado, USA

- Contributed to RouteE Compass, the Rust routing engine enabling Google Maps' eco-routing: TOML configuration injection, SI-unit energy model, and conda-forge packaging for OSX arm64.
- Reduced test-execution time for a 1 TB multi-threaded HPC data pipeline from 10 hours to 30 minutes (95%) by analyzing test-data dependencies, strategically truncating datasets, and parallelizing execution.
- Redesigned the visualization layer of the open-source em-public-dashboard: replaced pie charts with 100% stacked bar charts supporting labeled/inferred/sensed data with a unified color scheme — 30+ merged, maintainer-reviewed PRs.
- Built automated plot-testing workflows (pytest, Docker, GitHub Actions), cutting manual testing overhead ~60%.

Software Engineer Intern II May 2023 – Aug 2023
Yahoo Inc. Illinois, USA

- Owned end-to-end fuzzing research for production DNS infrastructure: learned and evaluated four engines (Google FuzzTest, LibFuzzer, AFL++, Honggfuzz) across the C++17/LLVM/Linux/Docker toolchain, identifying and reporting two critical memory-management vulnerabilities.
- Authored a ~50-page engineering document analyzing engine tradeoffs and enabling reproduction of the full setup; built CI pipelines running automated fuzzing 24/7.

Software Engineer I & II Jun 2018 – Jun 2022
Philips India Ltd. (Ultrasound R&D) Bangalore, India

- Sole developer of a multi-threaded, low-latency (5 ms) real-time audio module for the Lumify iOS app: owned requirements gathering with clinical scientists, framework selection (AVFoundation/AVAudioEngine), and delivery of the shipped feature in an App Store medical product.
- Engineered real-time audio-buffer processing end to end — tuning buffer sizes and sample rates against latency constraints — integrating a cross-platform C++ core with native Swift layers under an FDA-regulated release process.
- Implemented real-time diagnostic algorithms (Quad AFI, High Q Doppler) in C++ under 5 ms latency constraints, supporting the launch of a major new product line.
- Developed a secure multi-level user-management portal (C++, C#, WPF/MVVM); mentored new recruits across Agile projects.

Projects & Open Source

Energy Policy RAG Application github.com/iantei

- Built a Retrieval-Augmented Generation system (LangChain, ChromaDB, GPT-4o-mini) with a Cloud/Local LLM toggle, Polars/Parquet ETL, Docker GPU passthrough, and an 18-test pytest suite.

Open Source: contributor to NatLabRockies/routee-compass (Rust routing engine) and nanvix (Microsoft Research's Rust microkernel OS).

Education

Master of Computer Science, GPA 4.0 — Arizona State University, Tempe, AZ May 2024

B.Tech. Computer Science & Engineering — MNNIT, Prayagraj, India May 2018

Technical Skills

iOS & Mobile:	Swift, UIKit, AVFoundation / AVAudioEngine, MVVM, Xcode, cross-platform C++ integration, real-time audio
Languages:	Swift, C++, Python (Pandas, NumPy, Scikit-learn), Rust, C# (WPF), SQL
AI/LLM:	LangChain, RAG architecture, OpenAI API; daily use of Claude Code, Cursor, Copilot
Cloud & DevOps:	AWS (EC2, S3, SQS, Lambda), Docker, GitHub Actions, CI/CD, CMake, Linux
Databases:	PostgreSQL, MongoDB
Practices:	Automated testing, technical documentation, mentoring & onboarding, Agile, SDLC

Awards & Publications

- U.S. DOE-VTO AMC Team Award 2025 — RouteE & FASTSim enabled Google Maps' eco-routing feature.
- Co-author: Powell, B., Shrestha, A., et al., "Where are all the Electric Vans?", ASCE ICTD 2025.
- Philips Ultrasound R&D Hackathon: Winner 2019–20; Runner-up 2018–19; Team Up to Win 2022.